UNIVERSITY TECHNOLOGY MALAYSIA

LAB EXERCISE 2
SECJ1013-03
(PROGRAMMING TECHNIQUE I)

NO	NAME	MATRIC NO
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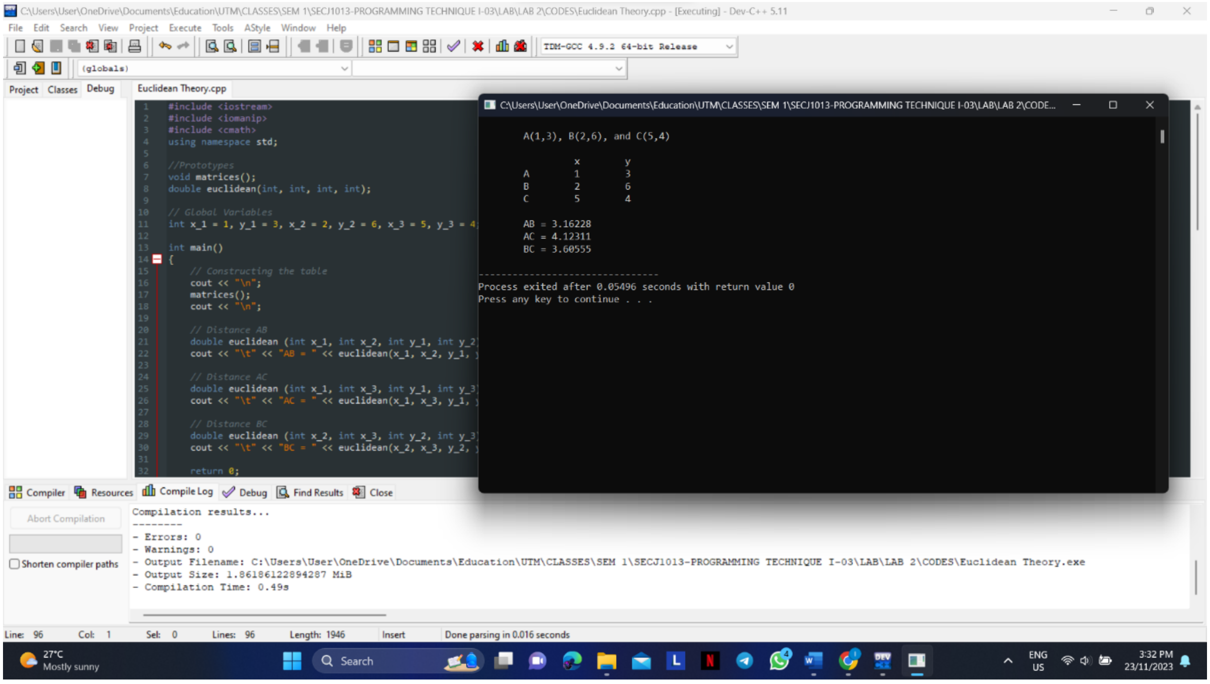
Code :

```
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

//Prototypes
void matrices();
double euclidean(int, int, int, int);

// Global Variables
int x_1 = 1, y_1 = 3, x_2 = 2, y_2 = 6, x_3 = 5, y_3 = 4;

int main()
{
	// Constructing the table
	cout << "\n";
	matrices();
	cout << "\n";

	// Distance AB
	double euclidean (int x_1, int x_2, int y_1, int y_2);
	cout << "\t" << "AB = " << euclidean(x_1, x_2, y_1, y_2) << "\n";

	// Distance AC
	double euclidean (int x_1, int x_3, int y_1, int y_3);
	cout << "\t" << "AC = " << euclidean(x_1, x_3, y_1, y_3) << "\n";

	// Distance BC
	double euclidean (int x_2, int x_3, int y_2, int y_3);
	cout << "\t" << "BC = " << euclidean(x_2, x_3, y_2, y_3) << "\n";

	return 0;
}

void matrices()
{
	int x [3] = {1, 2, 5};
	int y [3] = {3, 6, 4};
	char coordinate [3][3] = {"A", "B", "C"};

	for (int count = 0; count < 3; count++)
	{
		if (count == 0)
```

```
		{
			cout << "\t" << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
		}
		else if (count == 2 )
		{
			cout << ", and " << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
		}
		else
		{
			cout << ", " << coordinate[count] << "(" << x[count] << "," <<
y[count] << ")";
		}
	}

	cout << "\n";
	cout << "\n";

	cout << "\t" << " " << setw(9) << "x" << setw(9) << "y" << "\n";

	for (int count = 0; count < 3; count++)
	{
		cout << "\t" << coordinate[count] << setw(9) << x[count] << setw(9) <<
y[count] << "\n";
	}

}

double euclidean(int a, int b, int c, int d)
{
	// Difference of both x and y
	int difference_x = a - b;
	int difference_y = c - d;

	// The square of the difference of x and y
	int xpower2 = pow(difference_x,2);
	int ypower2 = pow(difference_y,2);

	// Square root of the sum of squares
	double distance = sqrt(xpower2 + ypower2);

	return distance;
}
```

```cpp
#include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;

//Prototypes
void matrices();
double euclidean(int, int, int, int);

// Global Variables
int x_1 = 1, y_1 = 3, x_2 = 2, y_2 = 6, x_3 = 5, y_3 = 4;

int main()
{
    // Constructing the table
    cout << "\n";
    matrices();
    cout << "\n";

    // Distance AB
    double euclidean (int x_1, int x_2, int y_1, int y_2);
    cout << "\t" << "AB = " << euclidean(x_1, x_2, y_1, y_2) << "\n";

    // Distance AC
    double euclidean (int x_1, int x_3, int y_1, int y_3);
    cout << "\t" << "AC = " << euclidean(x_1, x_3, y_1, y_3) << "\n";

    // Distance BC
    double euclidean (int x_2, int x_3, int y_2, int y_3);
    cout << "\t" << "BC = " << euclidean(x_2, x_3, y_2, y_3) << "\n";

    return 0;
}

void matrices()
{
    int x [3] = {1, 2, 5};
    int y [3] = {3, 6, 4};
    char coordinate [3][3] = {"A", "B", "C"};

    for (int count = 0; count < 3; count++)
    {
        if (count == 0)
        {
            cout << "\t" << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
        }
        else if (count == 2 )
        {
            cout << ", and " << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
        }
        else
        {
            cout << ", " << coordinate[count] << "(" << x[count] << "," << y[count] << ")";
        }
    }

    cout << "\n";
    cout << "\n";

    cout << "\t" << " " << setw(9) << "x" << setw(9) << "y" << "\n";

    for (int count = 0; count < 3; count++)
    {
        cout << "\t" << coordinate[count] << setw(9) << x[count] << setw(9) << y[count] << "\n";
    }

}

double euclidean(int a, int b, int c, int d)
{
    // Difference of both x and y
    int difference_x = a - b;
    int difference_y = c - d;

    // The square of the difference of x and y
    int xpower2 = pow(difference_x,2);
    int ypower2 = pow(difference_y,2);

    // Square root of the sum of squares
    double distance = sqrt(xpower2 + ypower2);

    return distance;
}
```

Output:


A(1,3), B(2,6), and C(5,4)
x y
A 1 3
B 2 6
C 5 4
AB = 3.16228
AC = 4.12311
BC = 3.60555
Process exited after 0.05496 seconds with return value 0
Press any key to continue . . .
Compilation results...
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\User\OneDrive\Documents\Education\UTM\CLASSES\SEM 1\SECJ1013-PROGRAMMING TECHNIQUE I-03\LAB\LAB 2\CODES\Euclidean Theory.exe
- Output Size: 1.86186122894287 MiB
- Compilation Time: 0.49s